

Lab 2 - Configuring Basic Single-Area OSPFv2

Answer Sheet

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Part 2: Configure and Verify OSPF Routing

Step 1: Verify OSPF neighbors and routing information.

b.

What command would you use to only see the OSPF routes in the routing table?

Route# show ip route O

Part 4: Configure OSPF Passive Interfaces

- g. Re-issue the **show ip route** and **show ip ospf neighbor** commands on R1 and R3, and look for a route to the 192.168.2.0/24 network.

What interface is R3 using to route to the 192.168.2.0/24 network? S0/0/0

S0/0/0

What is the accumulated cost metric for the 192.168.2.0/24 network on R3? 129

Does R2 show up as an OSPF neighbor on R1? Yes

Does R2 show up as an OSPF neighbor on R3? No

What does this information tell you?

R2 is advertising route information to R1 only and not to R3. Therefore, R3 can only route to 192.168.2.0/24 network through R1.

- h. Change interface S0/0/1 on R2 to allow it to advertise OSPF routes. Record the commands used below.

"R2(config-router)# no passive-interface s0/0/1"

- i. Re-issue the **show ip route** command on R3.

What interface is R3 using to route to the 192.168.2.0/24 network? s0/0/1

What is the accumulated cost metric for the 192.168.2.0/24 network on R3 now and how is this calculated?

_____ The accumulated cost is 65. It is calculated with the summation of the cost from R3 to R2 and R2 to 192.168.2.0/24 network. _____

Is R2 listed as an OSPF neighbor to R3? Yes

Part 5: Change OSPF Metrics

Step 1: Change the reference bandwidth on the routers.

- a. To reset the reference-bandwidth back to its default value, issue the **auto-cost reference-bandwidth 100** command on all three routers.

```
R1(config)# router ospf 1
```

```
R1(config-router)# auto-cost reference-bandwidth 100
```

```
% OSPF: Reference bandwidth is changed.
```

```
Please ensure reference bandwidth is consistent across all routers.
```

Why would you want to change the OSPF default reference-bandwidth?

_____ This is because the default value is much slower than most modern bandwidth which results in metric miscalculated as 1. Therefore, the value needs to be changed to a much higher value to compatible with the actual bandwidth. _____

Step 2: Change the bandwidth for an interface.

g.

Explain how the costs to the 192.168.3.0/24 and 192.168.23.0/30 networks from R1 were calculated.

_____ The cost to the 192.168.3.0/24 is the direct connection between R1's interface to R3's interface, the reference bandwidth is divided by the interface's bandwidth: $(100 \times 10^6) / (128 \times 10^3) = 8 = 781.25 = 782$. _____

The cost to 192.168.23.0/30 networks is between R1's interface to R2/R3's interfaces plus the cost from R2/R3's interfaces to 192.168.23.0/30 network. The cost between R2/R3's interfaces to the 192.168.23.0/30 network is calculated as around 64 so the total cost is $64 + 781.25 = 845.25 = 845$. _____

- i. Issue the **bandwidth 128** command on all remaining serial interfaces in the topology.

What is the new accumulated cost to the 192.168.23.0/24 network on R1? Why?

_____ The cost is 1562. This is because the cost from R2/R3'S interfaces to the 192.168.23.0/24 network were updated to 781 after updating the bandwidth to 128 Kb/s _____

Step 3: Change the route cost.

c.

Explain why the route to the 192.168.3.0/24 network on R1 is now going through R2?

___ This is because the R1's interface bandwidth to the 192.168.3.0/24 network is slower than the R2's interface to the network after the command. Now, the cost of R1 to R2 to the 192.168.3.0/24 network is 1563 which is lower than

~~1565~~

1566

Reflection

1. Why is it important to control the router ID assignment when using the OSPF protocol?

___ The router ID is for uniquely identify routers in the OSPF routing domain. Without any configuration, routers will pick the highest active IP addresses on any of their physical interfaces, which some of them might choose the same router ID and cause routing problems.

2. Why is the DR/BDR election process not a concern in this lab?

___ This is because each router is connected to each device with a point-to-point link. Only for a transient link, it will need a DR/BDR to represent itself.

3. Why would you want to set an OSPF interface to passive?

___ To avoid router advertising routing to passive interface, which can reduce traffic for LANs.

4. What aspect of OSPF makes the protocol hierarchical and permits the creation of very scalable networks?

___ The backbone area 0 of OSPF makes the protocol hierarchical and permits the creation of very scalable networks.

5. What single OSPF router configuration command allows the assignment of OSPF area 0 to all interfaces in the range 10.0.0.0 to 10.255.255.255?

___ " R1/2/3(config-router)# network 10.0.0.0 0.255.255.255 area 0"

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6. A router has three routes to the same network: one route from a RIP with a metric of 4, another from OSPF with a metric of 3053092, and another from EIGRP with a metric of 4039043. Which of the three routes will be used for the routing decision?

_____ The route from EIGRP since it has the lowest administrative distance and highest reliability.